Aircraft Variants for the 2027 Fly-off

Two complete aircraft, built from every dataset the team holds, and the reasoning behind each choice. The recommendation is written for the team to test, challenge and sign.

SITE TIMPA, TUCSON AZ
32.2647 N 111.2739 W
ELEV 667 M
DESIGN AIR 1.064 KG/M^3
FLY-OFF 15-18 APR 2027
CONFIG FREEZE 19 FEB 2027

VARIANTS IN THIS DOSSIER

A

Variant A: specimen name

Straight flapped high wing, one heavy sensor declared at tech.

MEAN SCORE 6.71
P(1ST) 0.42

B

Variant B: specimen name

Physically distinct hedge: winglet planform, different tail.

MEAN SCORE 6.58
P(1ST) 0.36

DOCUMENT	REVISION	PREPARED BY	STATUS
AUM-DBF-27-DB-002	A, 30 October 2026	Badr Obtel	Recommendation; team decision pending

01

Decision register

How to read this

1.1

Every decision in the dossier appears once in this register and once as a card in the body. The shaded column is left for the team.

Table 01

Specimen register.

ID	Decision	Recommendation	Grade	Changes if	Team
D-01	Wing area	1.0 m², straight, slotted flap	COMPUTED	Weigh-in 15 % heavy	
D-02	Declared sensor	At tech, 3–8 kg range	COMPUTED	Container fails TC-02	
D-03	Propulsion	8S single tractor	VERIFIED	Bench thrust below 95 N	

D-02 / EVIDENCE COMPUTED / CONFIDENCE HIGH / FREEZE AT TECH, APR 2027

D-02

Declare the sensor mass at technical inspection

Recommendation

Build one sensor body and one container that serve 3 to 8 kg; choose the declared maximum at tech, after the airframe is weighed and flown.

Why, plainly

The heavier the sensor we declare, the more the ground mission scores, but a declaration we cannot carry in Mission 2 scores nothing there. Choosing the number in April, when the real aircraft has been weighed and flown, removes most of that gamble.

The engineering

Across 120 paired draws at 0.9 m², freezing the unit now averages 6.30; choosing it at tech with what is known then averages 6.77, within 0.03 of perfect hindsight.

Derivation

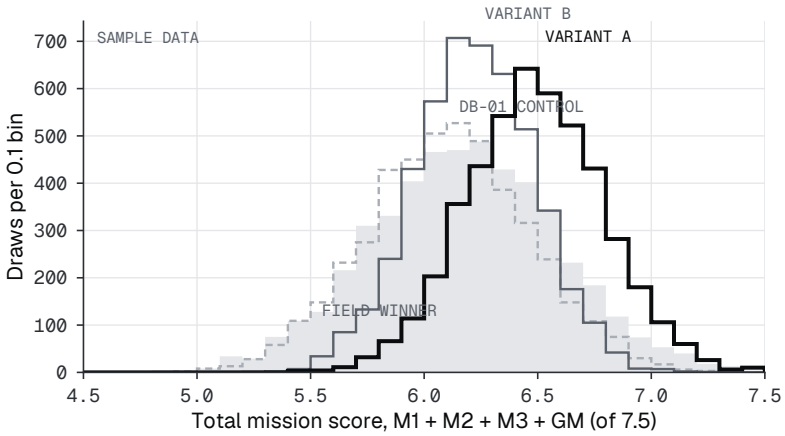
$$GM = 0.5 + \frac{W_s\, h}{\max(W_s h)}, \quad h \le 1.524 \text{ m}.$$ A 6 kg sensor gives $6 \times 1.524 = 9.14$ kg m, the expected-field anchor.

What would change it

The container cannot hold the g rating across the whole range (TC-02), or a Q&A ruling fixes the declared mass earlier.

DATA QUALITY / CFD, INDICATIVE

y+ averages 5.8 against a 1.5 target; the mesh fails three extra checkMesh tests from the relaxed layer pass; the two stalled angles end at the 1000-iteration cap with a 1.2–1.9 % force spread. Use for body and nacelle drag and moment trends, not for maximum lift.



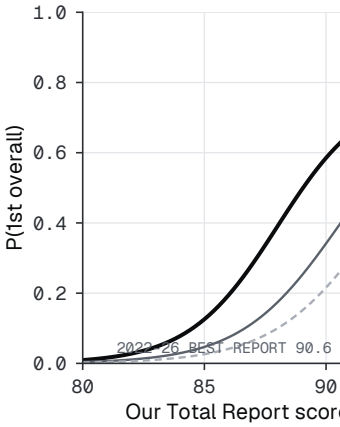


Figure 01

Specimen: the caption states the takeaway, not the topic.

04

Variant A

The recommended aircraft: what it is, why each part is the way it is, and how it scores against the field.

IN THIS CHAPTER

4.1 Specification

4.2 Decision cards

4.3 Aerodynamics and CFD

4.4 Stability and control

4.5 Missions and score

A Specimen variant name

Straight flapped high wing; conventional tail; fixed tricycle gear; single 8S tractor; closed-loop drum.

GENERAL ARRANGEMENT / TOP, SIDE, FRONT, ISO

THREE-VIEW PENDING

SPAN 1.820 M LENGTH 1.18 M SCALE 1:12.5 AT THIS SIZE CG MARKED AT M2 GROSS

PRINCIPAL CHARACTERISTICS

SPAN 1.820 M	WING AREA 1.00 M²	ASPECT RATIO 3.31	EMPTY MASS 6.4 KG
M2 GROSS 15.9 KG	STALL, M2 14.0 M/S	M2 SPEED 29.5 M/S	LAP, FLOWN 39.6 S
DECLARED SENSOR 3–8 KG	M3 LAPS 7	DROP HEIGHT 1.524 M	ENERGY, M2 71 WH
MEAN SCORE 6.71 / 7.5	P10 6.20	P(M2 AND M3) 0.81	P(1ST) 0.42

CONFIGURATION

WING	Straight, taper 0.7, SD7037	SENSOR	Tungsten dart, 3–8 kg
HIGH LIFT	Single-slotted flap, 60 % semispan	CONTAINER	PE foam, 381 mm envelope
TAIL	Conventional	D&R	Closed-loop drum, slip ring
GEAR	Fixed tricycle, faired	TOW LINE	2.84 m braided, 2 leads
PROPULSION	1 x AT7215 KV270, APC 20x15E	BAY	Two units, door
BATTERY	8S 3300 mAh, 97.7 Wh	STRUCTURE	Rib and spar, CF tube
FUSE	100 A blade	JOINT	Two-piece wing
RADIO	ExpressLRS 2.4 GHz, PWM	SHIPPING	Crate 1.2 x 0.5 x 0.4 m

MISSION PLAN

M1	Waived; credited by a successful Mission 2.	1.00
M2	One unit at the declared mass plus simulators as the weigh-in allows.	1.71
M3	Seven laps at 26.5 m/s with the ballast chosen per attempt.	3.00
GM	Declared mass from 1.524 m.	1.50